

# **CHAPTER: 7**

## **DATA DICTIONARY**

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## 7.1 Introduction

Data Dictionary is an important part of a project or system which contains all definition of elements in the system. In Data Dictionary you will find a list of all elements composing the data flowing through a system.

- The major elements of a system:
  - Data Flow
  - Data Store
  - Processes
- The data dictionary stores all details and description of these elements.
- The data dictionary provides additional information about the system.
- The data dictionary contains the data about data e.g. Student is a data and this student belongs to this course is a description of STUDENT data which is stored in data dictionary.
- Why is Data Dictionary important?
  - To manage the details in large system.
  - To communicate a common meaning for all system elements.
  - To document the features of system.
  - To determine where system changes should made.
  - To locate errors and omissions in the system.

## 7.2 List of Tables:

- **Table 7.2.1: user\_type**
  - **Description:** Stores user types of system (Admin/Organizer/Participant/Judge).

[Table 7.2.1: user\_types]

| Field Name   | Data type (size) | Constraint                  | Description                                  |
|--------------|------------------|-----------------------------|----------------------------------------------|
| user_type_id | BIGINT           | Primary key, Auto Increment | Unique role id                               |
| user_type    | Varchar(30)      | Not Null, Unique            | Role Name (ADMIN/ORGANIZER/PARTICIPANT/JUDE) |

- **Table 7.2.2: users**
  - **Description:** Stores login and basic profile details

[Table 7.2.2: users]

| Field Name | Data type (size) | Constraint                  | Description                               |
|------------|------------------|-----------------------------|-------------------------------------------|
| user_id    | BIGINT           | Primary key, Auto Increment | Stores the unique identifier of the user. |
| first_name | Varchar(100)     | Not Null                    | User's first name                         |
| last_name  | Varchar(100)     | Not Null                    | User's last name                          |
| email      | Varchar(150)     | Not Null, Unique            | User's email                              |
| password   | Varchar(255)     | Not Null                    | Encrypted password                        |
| role       | VARCHAR(30)      | NOT NULL                    | Role label                                |

|                 |              |                                  |                                                                          |
|-----------------|--------------|----------------------------------|--------------------------------------------------------------------------|
| profile_pic_url | VARCHAR(500) | NULL                             | Profile image url                                                        |
| active          | BOOLEAN      | NOT NULL                         | Account Status                                                           |
| otp             | VARCHAR(10)  | NULL                             | <ul style="list-style-type: none"> <li>OTP for password reset</li> </ul> |
| user_type_id    | BIGINT       | FK -><br>user_types.user_type_id | Role reference                                                           |

- **Table 7.2.3: user\_details**
  - **Description: Stores extended user profile information**

[Table 7.2.3: users\_details]

| Field Name      | Data type (size) | Constraint                  | Description           |
|-----------------|------------------|-----------------------------|-----------------------|
| user_detail_id  | BIGINT           | Primary key, Auto Increment | PK, Auto Increment    |
| user_id         | BIGINT           | FK -> users.user_id, UNIQUE | Owner user            |
| bio             | VARCHAR(1000)    | NULL                        | Short profile bio     |
| location        | VARCHAR(150)     | NULL                        | User location         |
| linkedin_url    | VARCHAR(255)     | NULL                        | LinkedIn Profile link |
| role            | VARCHAR(30)      | NOT NULL                    | Role label            |
| profile_pic_url | VARCHAR(500)     | NULL                        | Profile image url     |

- **Table 7.2.4: catagories**
  - **Description: Stores hackathon categories.**

**[Table 7.2.4: catagories]**

| <b>Field Name</b> | <b>Data type<br/>(size)</b> | <b>Constraint</b>  | <b>Description</b> |
|-------------------|-----------------------------|--------------------|--------------------|
| category_id       | BIGINT                      | PK, Auto Increment | Unique category id |
| category_nam      | VARCHAR(100)                | NOT NULL, UNIQUE   | Category name      |
| active            | BOOLEAN                     | NOT NULL           | Category status    |

- **Table 7.2.5: hackathon**
  - **Description: Stores hackathon categories.**

[Table 7.2.5: hackathon]

| Field Name              | Data type (size) | Constraint                   | Description                |
|-------------------------|------------------|------------------------------|----------------------------|
| hackathon_id            | BIGINT           | PK, Auto Increment           | Unique hackathon id        |
| title                   | VARCHAR(200)     | NOT NULL                     | Hackathon Title            |
| description             | TEXT             | NOT NULL                     | Event description          |
| status                  | VARCHAR(30)      | NOT NULL                     | UPCOMING/ONGOING/COMPLETED |
| event_type              | VARCHAR(30)      | NOT NULL                     | ONLINE/OFFLINE/HYBRID      |
| payment                 | VARCHAR(30)      | NOT NULL                     | FREE/PAID                  |
| location                | VARCHAR(200)     | NULL                         | Event location             |
| registration_start_date | DATE             | NOT NULL                     | Registration start         |
| registration_end_date   | DATE             | NOT NULL                     | Registration end           |
| min_team_size           | INT              | NOT NULL                     | Minimum team members       |
| max_team_size           | INT              | NOT NULL                     | Maximum team members       |
| user_id                 | BIGINT           | FK -> users.user_id          | Organizer id               |
| category_id             | BIGINT           | FK -> categories.category_id | Category id                |

- **Table 7.2.6: teams**
  - **Description: Stores team records for hackathons.**

**[Table 7.2.6: teams]**

| <b>Field Name</b> | <b>Data type<br/>(size)</b> | <b>Constraint</b>   | <b>Description</b>      |
|-------------------|-----------------------------|---------------------|-------------------------|
| team_id           | BIGINT                      | PK, Auto Increment  | Unique team id          |
| team_name         | VARCHAR(150)                | NOT NULL            | Team Name               |
| created_at        | DATETIME                    | NOT NULL            | Team creation timestamp |
| leader_user_id    | BIGINT                      | FK -> users.user_id | Team leader             |

- **Table 7.2.7: team\_members**

- **Description:** Stores members belonging to teams.

[Table 7.2.7: team\_members]

| Field Name     | Data type<br>(size) | Constraint          | Description          |
|----------------|---------------------|---------------------|----------------------|
| team_member_id | BIGINT              | PK, Auto Increment  | Unique member row id |
| team_id        | VARCHAR(150)        | FK -> teams.team_id | Team reference       |
| member_user_id | BIGINT              | FK -> users.user_id | Member user          |
| role           | VARCHAR(30)         | NOT NULL            | LEADER/MEMBER        |



- **Table 7.2.8: hackathon\_applications**
  - **Description:** Stores application submissions for hackathons.

[Table 7.2.8: team\_members]

| Field Name             | Data type<br>(size) | Constraint          | Description                    |
|------------------------|---------------------|---------------------|--------------------------------|
| application_id         | BIGINT              | PK, Auto Increment  | Unique application id          |
| team_id                | BIGINT              | FK -> teams.team_id | Team reference                 |
| participant_user_id    | BIGINT              | FK -> users.user_id | Applicant user                 |
| status                 | VARCHAR(30)         | NOT NULL            | PENDING/APPROVED/REJECTED/ etc |
| payment_status         | VARCHAR(30)         | NOT NULL            | WAIVED/PENDING/PAID            |
| applied_at             | DATETIME            | NOT NULL            | Application timestamp          |
| submission_url         | VARCHAR(500)        | NULL                | Project submission URL         |
| frontend_github_link   | VARCHAR(500)        | NULL                | Frontend repo URL              |
| backend_github_link    | VARCHAR(500)        | NULL                | Backend repo URL               |
| submission_description | TEXT                | NULL                | Submission details             |

- **Table 7.2.9: judge\_assignments**
  - **Description:** Stores team records for hackathons.

[Table 7.2.9: judge\_assignments]

| Field Name          | Data type<br>(size) | Constraint                   | Description                  |
|---------------------|---------------------|------------------------------|------------------------------|
| judge_assignment_id | BIGINT              | PK, Auto Increment           | Unique assignment id         |
| hackathon_id        | BIGINT              | FK -> hackathon.hackathon_id | Hackathon reference          |
| judge_user_id       | BIGINT              | FK -> users.user_id          | Assigned judge               |
| assigned_by_user_id | BIGINT              | FK -> users.user_id          | Organizer/Admin who assigned |
| assigned_at         | DATETIME            | NOT NULL                     | Assignment timestamp         |

- **Table 7.2.10: judge\_scores**
  - **Description:** Stores scorecards given by judges.

[Table 7.2.10: judge\_scores]

| Field Name     | Data type<br>(size) | Constraint                                                 | Description           |
|----------------|---------------------|------------------------------------------------------------|-----------------------|
| judge_score_id | BIGINT              | PK, Auto Increment                                         | Unique score id       |
| application_id | BIGINT              | BIGINT  <br>FK -><br>hackathon_applications.application_id | Application reference |
| judge_id       | BIGINT              | FK -><br>users.user_id                                     | Judge Reference       |
| score          | DECIMAL(5,2)        | NOT NULL                                                   | Numeric score         |
| remarks        | VARCHAR(1000)       | NULL                                                       | Judge comments        |